ADVANCED COMPUTER NETWORKS · TEAM 7 · MID-PROJECT EVALUATION

Streaming a live lecture over two weak internet connections

Send whole video frames, not scattered packets, over two paths at once

Multipath QUIC · emulated cellular links · 2 November 2026

A teacher's live video needs a steady upload that one phone often cannot give

1 · THE PROBLEM

2.6 Mbit/s

steady upload needed for a 720p class video (Zoom's figure)

31–44%

of mobile tests in India miss a consistent-quality bar, though average uploads (5.9–8.4 Mbit/s) look fine

2 links

are often available: WiFi + mobile data, or two phones

Sources: Zoom bandwidth requirements; Opensignal India report, Feb 2026 (56–69% of tests on the three private operators met its bar).

Multipath protocols split the video packet by packet

2 · WHAT EXISTS TODAY

Multipath QUIC can already use two links in one connection. A **scheduler** picks a link for every packet:

- MinRTT: the faster link first
- round-robin: turn by turn
- redundant: both links
- ECF: by expected completion (Lim et al., 2017)

Commercial bonding (LiveU) is closed; research that sends whole frames (MoMQ, 2026) works at a relay, not at the teacher's phone.

The weak spot

Packets of one frame land on both links. A frame can be shown only when its **last** packet arrives, so the slow link decides when every frame appears.

Treat each video frame as one unit and place it where it finishes first

3 · OUR IDEA

1 Predict

For each new frame, estimate when it would finish on each link.

2 Place or split

Send the whole frame on the earlier link; split a large frame across both only if that finishes sooner.

3 Rescue

If a link stalls (no progress for one timeout), resend its unfinished part on the other link.

Built in the sender on top of multipath QUIC; each frame part is its own QUIC stream. The parts are known ideas (ECF, DEMS, stream-aware MPQUIC); the new part is doing it at the teacher's uplink and measuring it with deadlines and decoding rules.

A frame's finish time is predictable, so we can balance it

Finish time of a frame part of S bytes on link p:

$$T_p = \underbrace{\frac{U_p + S}{R_p}}_{\text{drain the queue}} + \underbrace{(\mathrm{sRTT}_p - \mathrm{minRTT}_p)}_{\text{wait in the buffer}} + \underbrace{\frac{\mathrm{minRTT}_p}{2}}_{\text{fly one way}}$$

U_p: bytes already queued on p; R_p: measured delivery rate. Write d_p for the two delay terms.

Split so both parts finish together:
$T_1(S_1) = T_2(S - S_1)$ gives

$$S_1 = \frac{\dfrac{U_2 + S}{R_2} + d_2 - \dfrac{U_1}{R_1} - d_1}{\dfrac{1}{R_1} + \dfrac{1}{R_2}}, \qquad 0 \le S_1 \le S.$$

Worked example (formula only)

$R_1 = R_2 = 200$ KB/s, $d_1 = 15$ ms, $d_2 = 75$ ms, empty queues, $S = 85$ KB:
$S_1 = 0.485\,\mathrm{s}/0.01\,\mathrm{s/KB} = 48.5$ KB.
Both parts done at **258 ms**; one link alone: 440 ms.

Same video, same links, every scheduler: measured, not assumed

5 · HOW WE TEST IT

Testbed

Two emulated links on one laptop: delay, speed, 200 ms buffer and outages set exactly; later, real two-operator recordings replayed.

Video

A real 60-s NPTEL blackboard lecture, encoded like a live stream: 2.6 Mbit/s, 25 frames/s.

Score

Frame i is shown iff $L_i \le D = 250$ ms, where $L_i = \max_j c_j - t_i$ over i and the frames it depends on. Freeze: a gap ≥ 190 ms.

Five predictions, written down before any run

One link is often not enough (H1) · today's schedulers waste capacity when delays differ (H2) · our scheduler freezes less (H3) · it needs no special protocol (H4) · it helps most with big keyframes (H5).

8 paired runs per comparison: Wilcoxon's smallest $p = 2/2^8 = 0.0078 <$ Holm's first threshold $0.05/6 = 0.0083$.

The measurement setup is built and verified

6 · BUILT AND CHECKED SO FAR

Emulator is accurate

Delay within 0.4 ms, speed within 0.2%, buffer 200 ± 1 ms. Checking caught two emulator bugs, now fixed.

Protocol fixes

Four small changes to the QUIC library (TQUIC), each with a test; one fixes a connection stall in the library itself.

Multipath works

20 of 20 test transfers used both links (each 49–51% of the data).

Pipeline

Video streamer, one-command runs, scoring and statistics code, all with tests.

Plan to the final demo on 17 November

8 · NEXT STEPS

By 11 Nov

Ideal-link reference; real recordings at 3 sites. Decide: is there a problem worth solving?

12–14 Nov

Build the frame scheduler; run the full comparison ($\sim$1,240 runs, overnight).

15–17 Nov

Statistics, plots, report; final demo on 17 Nov.

Main risks

Real recordings need two phones on two operators and a cloud server (tools ready and tested). The schedule is tight: about 25 hours of runs in four nights.

Thank you. Questions?

Problem: one weak uplink. Existing fix: split packets over two links.
Our bet: split by whole frames, timed to finish together.

Video: Prof. V. Balakrishnan, IIT Madras, NPTEL (CC BY-SA). Stack: TQUIC (Tencent), Linux netem/htb, ffmpeg/x264.